Lecture 31 (*May 10*)

Reading: Section 12.2
Viewing: none

Let us generalize the previous setup to an arbitrary number of random variables.

Definition 28. *Given n random variables $X_1, \ldots, X_n$, their covariance matrix is:*

$$K = \begin{bmatrix} \Sigma_{X_1X_1} & \cdots & \Sigma_{X_1X_n} \\ \vdots & \ddots & \vdots \\ \Sigma_{X_nX_1} & \cdots & \Sigma_{X_nX_n} \end{bmatrix} \tag{288}$$

The covariance matrix is a useful tool for determining how (un)correlated a collection of random variables is, for example if $X_1 =$ age, $X_2 =$ height, $X_3 =$ weight among a population. In fact, it is often convenient to put all of the random variables in a vector, called a **random vector**:

$$\boldsymbol{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \tag{289}$$

Let $p_1, p_2, \ldots$ to be the probability that $\boldsymbol{X}$ takes a certain number of vector values $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots,$ where $\boldsymbol{x}_i = \begin{bmatrix} x_{1,i} \\ \vdots \\ x_{n,i} \end{bmatrix}$. In the setting of two random variables in (278), the probabilities in question were denoted by $p_{ij} = \text{Prob}\left(\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} x_i \\ y_j \end{bmatrix}\right)$. Then the mean of the random vector (289) is:

$$\boldsymbol{\mu} = \sum_i p_i \boldsymbol{x}_i$$

(note that $\boldsymbol{\mu}$ is a vector) and the covariance matrix is:

$$K = \sum_i p_i \begin{bmatrix} (x_{1,i} - \mu_1)^2 & \cdots & (x_{1,i} - \mu_1)(x_{n,i} - \mu_n) \\ \vdots & \ddots & \vdots \\ (x_{1,i} - \mu_1)(x_{n,i} - \mu_n) & \cdots & (x_{n,i} - \mu_n)^2 \end{bmatrix} =$$

$$= \sum_i p_i \begin{bmatrix} x_{1,i} - \mu_1 \\ \vdots \\ x_{n,i} - \mu_n \end{bmatrix} \begin{bmatrix} x_{1,i} - \mu_1 \\ \vdots \\ x_{n,i} - \mu_n \end{bmatrix}^T = \sum_i p_i (\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^T = E\Big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T\Big] \tag{290}$$

Apart from the fact that vector and matrix notation gives the neat formula (290) for the covariance matrix, the equality above proves that the covariance matrix of any number of random variables is positive semidefinite. Moreover, any linear combination of the random variables $X_1, \ldots, X_n$:

$$X = \boldsymbol{c}^T \boldsymbol{X} \qquad \text{for some} \qquad \boldsymbol{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is a random variable (for example, in the age–height–weight example, X could be $2 \cdot \text{age} - 3 \cdot \text{height} + 7 \cdot \text{weight}$). The variance of X can be recovered from the covariance matrix K as:

$$E\Big[(X-\mu)^2\Big] = E\Big[(X-\mu)(X-\mu)^T\Big] = E\Big[(\boldsymbol{c}^T\boldsymbol{X} - \boldsymbol{c}^T\boldsymbol{\mu})(\boldsymbol{c}^T\boldsymbol{X} - \boldsymbol{c}^T\boldsymbol{\mu})^T\Big] =$$
$$= E\Big[(\boldsymbol{c}^T\boldsymbol{X} - \boldsymbol{c}^T\boldsymbol{\mu})(\boldsymbol{X}^T\boldsymbol{c} - \boldsymbol{\mu}^T\boldsymbol{c})\Big] = E\Big[\boldsymbol{c}^T(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X}^T - \boldsymbol{\mu}^T)\boldsymbol{c}\Big] = \boldsymbol{c}^T K \boldsymbol{c} \quad (291)$$

which is precisely the energy of the vector $\boldsymbol{c}$ with respect to the positive semidefinite covariance matrix K. The matrix K is positive definite unless the energy is 0 for a non-zero vector $\boldsymbol{c}$. Hence:

the covariance matrix is positive definite, unless

the random variables $X_1, \ldots, X_n$ are linearly dependent

where "linearly dependent" means that there exists some linear combination of $X_1, \ldots, X_n$ which is constant (since the constant random variables are precisely those with 0 variance).

Example 10. *Here's a quick application of this matrix formalism. Suppose you have random variables X and Y, with means μ and ν, and variances Σ_{XX} and Σ_{YY}. What are the mean and variance of the random variable $Z = X + Y$? You can compute these by writing:*

$$Z = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

The mean of Z is just:

$$\rho = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \nu \end{bmatrix} = \mu + \nu$$

which is perhaps not so surprising. However, the variance of Z is given by formula (291)*:*

$$\Sigma_{ZZ} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{XY} & \Sigma_{YY} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \Sigma_{XX} + \Sigma_{YY} + 2\Sigma_{XY}$$

Thus, the variance of a sum of random variables involves not only the sum of the individual variances, but also a cross-term involving the covariance of the two variables.

In this language, we can ask what is the meaning of diagonalizing the covariance matrix:

$$K = QDQ^T \quad (292)$$

where D is a diagonal matrix with non-negative eigenvalues, and Q has orthonormal columns. This procedure is called **principal component analysis (PCA)** in probability and statistics. Mathematically, this entails defining the following random vector:

$$\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \boldsymbol{Y} := Q^T\boldsymbol{X} \qquad \text{whose mean will be} \qquad \boldsymbol{\nu} = Q^T\boldsymbol{\mu} \quad (293)$$

This simply means that the entries $Y_1, \ldots, Y_n$ of $\boldsymbol{Y}$ are all random variables, given by certain linear combinations of the random variables that make up $\boldsymbol{X}$. In this case, the covariance matrix corresponding to the random vector $\boldsymbol{Y}$ is:

$$E\Big[(\boldsymbol{Y}-\boldsymbol{\nu})(\boldsymbol{Y}-\boldsymbol{\nu})^T\Big] = E\Big[Q^T(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^T Q\Big] = Q^T K Q = D = \begin{bmatrix} \Sigma_1 & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \Sigma_n \end{bmatrix}$$

Since D is a diagonal matrix, this implies that the random variables $Y_1, \ldots, Y_n$ are mutually uncorrelated (since their pairwise covariances are 0), and that Σ_i is the variance of Y_i. Therefore, the diagonalization (292) corresponds to finding n linear combinations of the random variables $X_1, \ldots, X_n$, denoted by $Y_1, \ldots, Y_n$ above, which are pairwise uncorrelated random variables.